

Chapter 3. Volatility Models

Financial Engineering

SHI, Chao (史钊)

Shanghai University of Finance and Economics



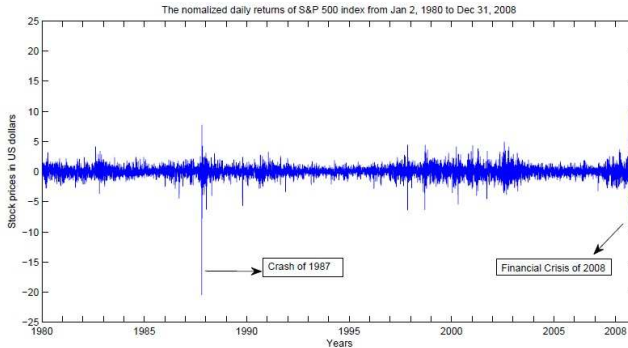
Content

- 1 **Flaws of the BSM**
 - Leptokurtic Feature
 - Implied Volatility Curve
- 2 **Stochastic Volatility Models**
 - Local Volatility Model
 - Stochastic Volatility Models
 - Affine Models
- 3 **Models with Jumps**
 - Jump-Diffusion Models
 - Exponential Lévy Models
 - Subordinator

Black-Scholes Model: Advantages and Flaws

- A “good” financial dynamic model...
 - (1) closely fits the features of empirical market data;
 - (2) is compact and elegant in mathematics;
 - (3) provides option prices in closed-form, or easy to compute via computational procedures.
- The BSM performed very well, and is still applied frequently now, but...
 - (1) cannot capture the leptokurtic (尖峰厚尾) feature of asset prices, which widely appears in real market;
 - (2) cannot indicate the varying implied volatility (IV, 隐含波动率) of options.

The Leptokurtic Feature



$$P(Z < -21.1550) \approx 1.4 \times 10^{-107}.$$

The Implied Volatility

- Under the BSM, European option price is given by the Black-Scholes option pricing formula:

$$V_0 = S_0 \cdot N(d_1) - e^{-rT} K \cdot N(d_2),$$

where

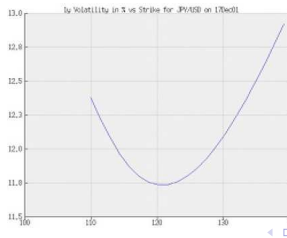
$$d_1 := \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma \sqrt{T}},$$

$$d_2 := d_1 - \sigma \sqrt{T} = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma \sqrt{T}}.$$

- Therefore, option price can be considered as a function $V(\sigma)$ of the latent (不可观测) variable σ .

Non-constant Implied Volatility

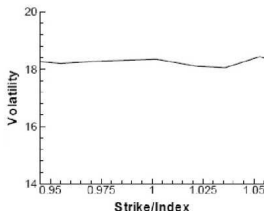
- Options have market quotes. Then we solve $V(\sigma) = V_{\text{market}}$ and obtain the solution $\tilde{\sigma}$, which is known as the **implied volatility (IV, 隐含波动率)**.
- Different option contracts provide different $\tilde{\sigma}$, containing profound market information.
- $\tilde{\sigma}$ is often not constant...



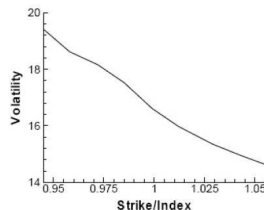
Before and after the “Black Monday”

The IV curve became more unstable after the “Black Monday” in 1987.

Representative S&P 500 implied volatilities prior to 1987

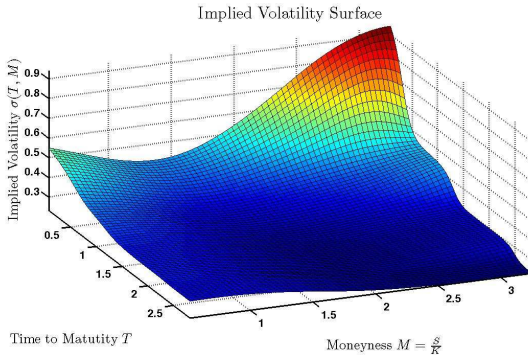


after 1987



The IV Surface

The IV surface plots IVs for different T and K/S_0



The volatility Smile

- IV curve is often convex with respect to K/S_0 . This phenomenon is the well-known **volatility smile**.
- Possible reasons
 - The BSM is too ideal
 - Prices of Deep ITM and OTM options are often higher than theoretical prices, which can be regarded as automatic correction for such problem by the market.
 - Contains market risk information.

The World beyond the BSM

- After the flash crash in 1987, 2008, and 2020, both the industry and the academia struggled to improve financial dynamic models which capture the market features better.
 - Let the volatility be a function.
 - Let the volatility be a diffusion process.
 - Multidimensional SDE
 - Allow jumps in the sample paths.
- New models can relieve the flaws of the BSM.

Local Volatility Model

- The first generation of volatility model sets the volatility as a function, and is thus called as **local volatility model** (局部波动率模型):

$$\frac{dS_t}{S_t} = (r_t - q_t)dt + \sigma(S_t, t; s_0)dW_t.$$

- European option prices under the local volatility model follow a Fokker-Plank forward equation:

$$\left(\frac{\partial}{\partial T} - L_{T,Y}^* \right) \phi(S, t; Y, T) = 0,$$

where

$$L_{T,Y}^* \phi := \frac{1}{2} \frac{\partial^2}{\partial Y^2} (\sigma^2(S_t, t) Y^2 \phi(S_t, t)) - \frac{\partial}{\partial Y} (\mu Y \phi(S_t, t)).$$

European Option Pricing*

- Derive along the Fokker-Plank forward equation, finally one obtains the Dupire equation:

$$\frac{\partial C}{\partial T} = \frac{1}{2}\sigma^2 K^2 \frac{\partial^2 C}{\partial K^2} + \mu_T(C - K) \frac{\partial C}{\partial K} - r_T C.$$

- Set

$$C(S_0, K, T) = C_{BS}(S_0, K, \sigma_{BS}(S_0, K, T), T).$$

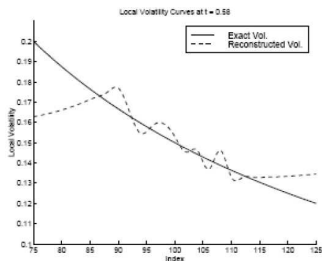
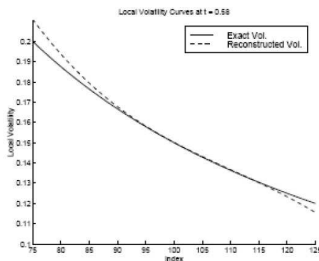
Here, σ_{BS} can be regarded as a “substitution of constant volatility”.

- Then the option prices can be expressed similarly with the Black-Scholes pricing formula.

How to Determine the Volatility Function*

Coleman, Li and Verma (1997) designs a two-step calibration (校准):

- Bivariate spline interpolation (二元三次样条差值)
- Asymptotic calibration



Advantages and Flaws

- Advantages

- Produces the volatility smile.
- Provide accurate prices for even exotic options.

- Shortcomings

- Option prices in the reality are only available for a little batch of the combination $\{K, T\}$.
- Numerical differentiation is unstable. For second-order differentiations, numerical algorithm is extremely unstable, even useless.
- Needs frequent calibration.

Stochastic Volatility Models

- The market quickly found flaws of local volatility model. Some trading business called for better dynamic models.
- Then the second generation of volatility model appeared soon. It is the stochastic volatility (SV) model, which introduced a latent stochastic process to describe the volatilities.
- The principle of SV involves risk premium in financial econometrics, and HJB equation in control theory.

The Heston SV Models

- Stein and Stein (1991, RFS) conducts concrete empirical studies, and suggests that the volatility follows an Ornstein-Uhlenbeck process:

$$d\sigma_t = -\alpha_t\sigma_t dt + \beta_t dW_t.$$

- Later, Heston (1993, RFS) follows such suggestion and proposes the [Heston SV model](#):

$$\left\{ \begin{array}{l} dS_t = \mu_t S_t dt + \sigma_t S_t dW_t, \\ d\sigma_t = -\alpha_t \sigma_t dt + \beta_t dZ_t, \\ dW_t dZ_t = \rho_t dt. \end{array} \right.$$

An Alternative Form

- Define a variance process $V_t := \sigma_t^2$.
- Replace the volatility with the variance via Itô's Lemma:

$$\begin{cases} dS_t = \mu_t S_t dt + \sqrt{V_t} S_t dW_t, \\ dV_t = \kappa(\theta - V_t)dt + \sigma \sqrt{V_t} dZ_t, \\ dW_t dZ_t = \rho_t dt. \end{cases}$$

This form is more common.

- Here, the variance process $\{V_t\}$ is governed by a CIR model, which possesses a mean-reverting property.

The Return Process

- For the price process $\{S_t\}$ of the underlying asset, define a **return process (回报过程)** by

$$X_t := \ln \left(\frac{S_t}{S_0} \right).$$

- The return process $\{X_t\}$ indicates the dynamic return of an asset, and is very useful in derivative pricing schemes.
- The **moment generating function (MGF, 矩生成函数)** of the marginal distribution X_t is denoted as:

$$\phi(X_t, V(t), t; z) := E \left[e^{zX_t} \mid \mathcal{F}_t \right], \quad z \in \mathbb{C}.$$

Option Pricing under the Heston SV Model*

- The function $\phi(X_t, V(t), t; z)$ above satisfies the Kolmogorov backward equation:

$$\begin{cases} \frac{\partial \phi}{\partial t} + \kappa(\theta - V)\frac{\partial \phi}{\partial V} - \frac{1}{2}V\frac{\partial \phi}{\partial X} + \frac{1}{2}\sigma^2 V\frac{\partial^2 \phi}{\partial V^2} + \sigma\rho V\frac{\partial^2 \phi}{\partial V \partial X} + \frac{1}{2}V\frac{\partial^2 \phi}{\partial X^2} = 0, \\ \phi(x, V, T; z) = e^{zx}. \end{cases}$$

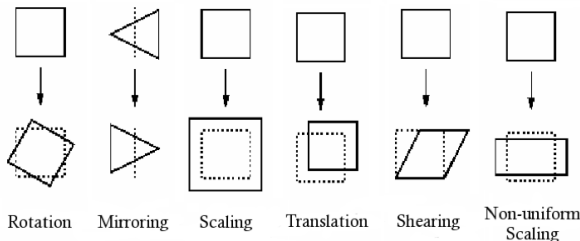
- Heston (1993, RFS) **guesses** that the solution has a form

$$\phi(x, v, \tau; z) = e^{A(\tau, z) + B(\tau, z)V + zx},$$

where $\tau := T - t$. Then solve a two-dimensional ODE with respect to two functions A and B , to obtain a specific form of the function ϕ .

Affine Models

- Affine (仿射) transformation:

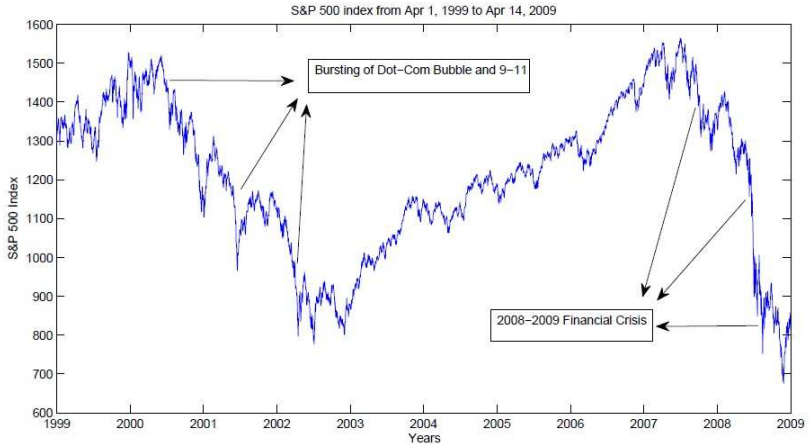


- A multidimensional diffusion process

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t$$

is defined as an **affine model**, if both $\mu(x, t)$ and $\sigma(x, t)\sigma^\top(x, t)$ are linear functions in x .

Discontinuous Stock Prices



Counting Process

- Counting process (计数过程) $\{N(t) : t \geq 0\}$ is a continuous-time stochastic process, expressing the number of times that a special incident happens up to time t .
- The values are nonnegative integers.
- Monotonically increasing
- Recall that for a stochastic process $\{X_t\}$, any random variable $X_t - X_s$ (here $t \geq s$) is defined as an increment (增量) of $\{X_t\}$.

Poisson Process

A counting process $\{N(t) : t \geq 0\}$ is defined as a **Poisson process (泊松过程)** with a parameter λ , if

- 1. $N(0) = 0$;
- 2. $\{N(t)\}$ is a process with independent increments;
- 3. $\{N(t)\}$ is a process with stationary increments;, and $N_t - N_s \sim Pn(\lambda(t - s))$, i.e.,

$$P(N_t - N_s = n) = \frac{[\lambda(t - s)]^n}{n!} e^{-\lambda(t-s)}, \quad \text{for all } n \in \mathbb{N}.$$

Compound Poisson Process

- $Y_1, Y_2, \dots$ is a sequence of i.i.d. random variables, and each term has a distribution F_Y . $\{N(t) : t \geq 0\}$ is a Poisson process with a parameter λ . Then the stochastic process

$$X(t) := \sum_{i=1}^{N(t)} Y_i$$

is known as a **compound Poisson process** with a parameter λ and the distribution F_Y .

- Randomization on the jump sizes
- Different distribution F_Y leads to different model $X(t)$.

Jump-Diffusion Models

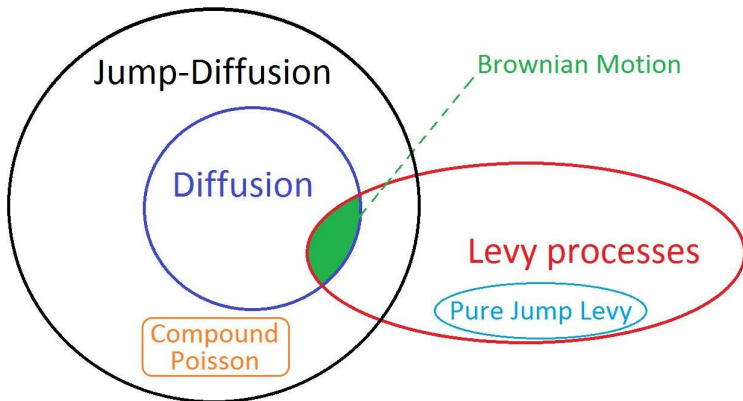
- Extreme financial events cause fierce fluctuations in asset prices, which enlightens us to release the continuous property of diffusion processes.
- **Jump-diffusion process (跳-扩散过程)**

$$dX_t = \mu_t dt + \sigma_t dW_t + d\left(\sum_{i=1}^{N_t} Y_i\right),$$

is often imposed on the return process, where $\{N_t\}$ is usually assumed to be a Poisson process, and Y_i 's indicate the jump sizes.

- Capture the instability of financial system, and characterize the volatility smile; see Kou (2002, MS).

Relationship of All Models



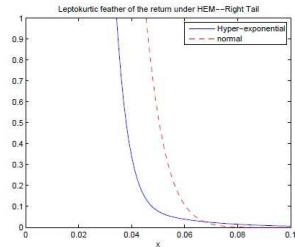
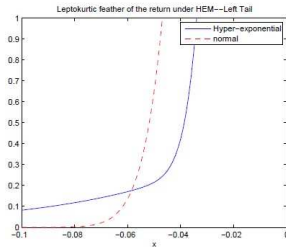
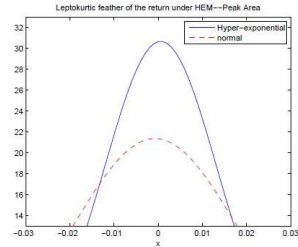
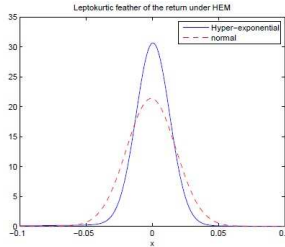
Merton's model and Kou's model

- The Merton's model assumes Y_i to be normal, which make the marginal distribution and option price easy to obtain.
- However, path-dependent option pricing becomes very difficult under Merton's jump-diffusion model.
- Instead, Kou (2002, MS) assumes that Y_i has the PDF below:

$$f_Y(x) = p\lambda e^{-\lambda x} \mathbf{1}_{\{x \geq 0\}} + (1-p)\mu e^{\mu x} \mathbf{1}_{\{x < 0\}}.$$

This special double-exponential structure provides analytical solutions for most path-dependent options.

Comparison of the Tails



Exponential Lévy Models

- Another broad category of dynamic models containing jumps is the celebrated exponential Lévy model. It is not provided by SDE.
- The exponential Lévy model assumes that the return process $\{X_t\}$ follows a **Lévy process**:
 - starting from 0;
 - independent and stationary increments;
 - some weak continuousness.
- Each Lévy process can be described by a Laplace exponent, which is very useful in option pricing.
- Examples: Kou's model; Variance-Gamma model

Laplace Exponent

- Due to the independent and stationary increments of a Lévy process $\{X_t\}$, we decompose

$$X_t = X_{\frac{1}{n}t} + (X_{\frac{2}{n}t} - X_{\frac{1}{n}t}) + \cdots + (X_t - X_{\frac{n-1}{n}t}),$$

then take limit and obtain

$$E[e^{sX_t}] = e^{\Psi(s)t}.$$

The function $\Psi(s)$ is just the **Laplace exponent** of the Lévy process $\{X_t\}$.

- Laplace exponent fully determines the class of FDDs of a Lévy process.

Examples of Laplace Exponents

- Example 1: Poisson process

$$\Psi(s) = \lambda(1 - e^s).$$

- Example 2: Brownian motion

$$\Psi(s) = s^2/2.$$

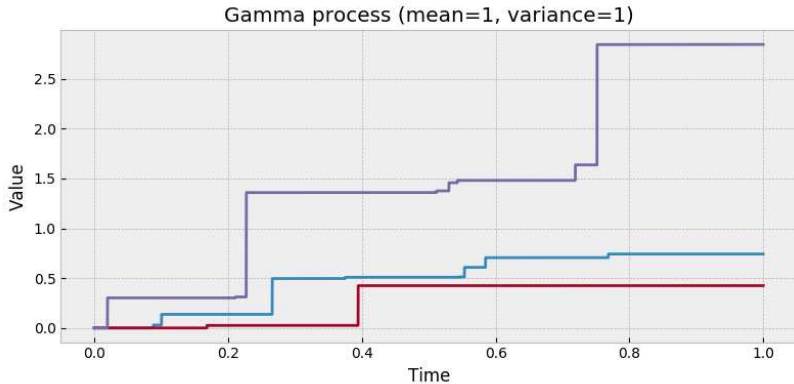
- Example 3: Gamma process (has monotonically increasing sample paths)

$$\Psi(s) = -\gamma \log(1 - s/\lambda), \quad s < \lambda.$$

- Example 4: Variance-Gamma process

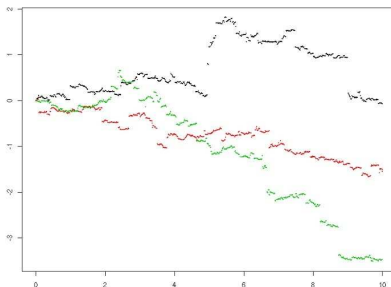
$$\Psi(s) = \frac{1}{\beta} \log \left(\frac{1}{1 - b\beta s + \sigma^2 \beta s^2 / 2} \right).$$

Gamma Process

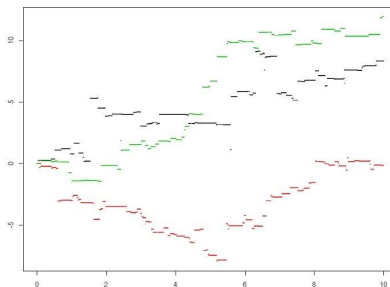


Pure Jump Lévy Processes

Normal-inversed Gaussian model



Variance-Gamma model



Variance-Gamma process can be defined as the difference of two different gamma processes, or equivalently, defined by its Laplace exponent.

Interesting Properties of Pure Jump Lévy Processes

- Only jump, without drift and diffusion
- Jump for **infinite times** in any arbitrary small interval of the time
- “Speed density” of all jump sizes of the Variance-Gamma process:

$$f(x) = \frac{1}{\kappa|x|} e^{Ax - B|x|}.$$

Meaning: the jump size can be any real value, and those jump sizes near 0 have unboundedly large intensities.

An Alternative Characterization of Lévy Processes*

- Enlightened by the “speed density” above of a Variance-Gamma model, we can specify a Lévy process more directly than adopting Laplace exponent.
- **Lévy-Itô Decomposition Theorem**: Any Lévy process can be decomposed into the sum of a pure drift, a pure diffusion, and a pure jump.
- The triplet made up by the drift, the diffusion, and the speed density of jump sizes is known as the **Lévy triplet**, which can determine a Lévy process.

Subordinator*

- Subordinator is a positive-valued stochastic process whose sample paths are all monotonically increasing. It can stand for a time change.
- Idea: by using the subordinator to replace the time of a stochastic process, we obtain a new stochastic model as a compound mapping.
- calendar time $\rightarrow$ subjective time
- Replace the time of a Brownian motion with a Gamma process $\gamma(t)$. Then we construct a Variance-Gamma model again, by the fourth kind of methods.

$$VG(t; \beta, b, \sigma) := W(\gamma(t; 1, \beta); b, \sigma).$$